

The background of the slide is a solid light pink color. Scattered across this background are several slices of lemons, cut into wedges. Each slice is positioned to cast a soft, dark pink shadow onto the surface below it, creating a sense of depth and texture. The lemon slices are arranged in a non-uniform pattern, with some in the foreground and others in the background.

15C+16 CHEMICAL EQUILIBRIUM

UNIT 02: HOW REACTIONS HAPPEN

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 15

1. Define and explain the condition of equilibrium.
2. Determine equilibrium constant expressions and their values.
3. Use equilibrium constant expressions as formulas.
4. Manipulate equilibrium constant expressions based on changes to the chemical equation.
5. Define and explain the reaction quotient.
6. Predict the direction of a reaction using the relationship between the reaction quotient and the equilibrium constant.
7. Use ICE tables to solve equilibrium problems.
8. Identify the reaction conditions that shift the equilibrium position of a reaction.

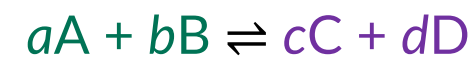
CHAPTER 16

1. Write ionization reactions for strong and weak acids and bases.
2. Calculate the hydrogen ion concentration in solutions of strong acids and the hydroxide ion concentration in solutions of strong bases.
3. Identify conjugate acid-base pairs using the Brønsted–Lowry definitions of acids and bases.
4. Interpret the relationship between K_a or K_b on the concentrations of species in acidic or basic solutions.
5. Calculate the pH of a solution.

Le Châtelier's Principle

15.6

When a system is at equilibrium and the conditions that affect the equilibrium are changed (perturbed), the rates of the forward and reverse reactions may be affected *differently*.



$$K = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

If $\text{Rate}_{\text{for}} \neq \text{Rate}_{\text{rev}}$, more reactants or more products will be produced until the *correct* ratio of products to reactants, governed by K , is re-established.

Systems will always tend toward equilibrium (K)!

Le Châtelier's Principle

- If a *stress* is applied to a system at equilibrium, the reaction will tend to *shift* in a direction to relieve the stress and re-establish equilibrium.
- Le Châtelier's principle is based on the comparison of Q to K .

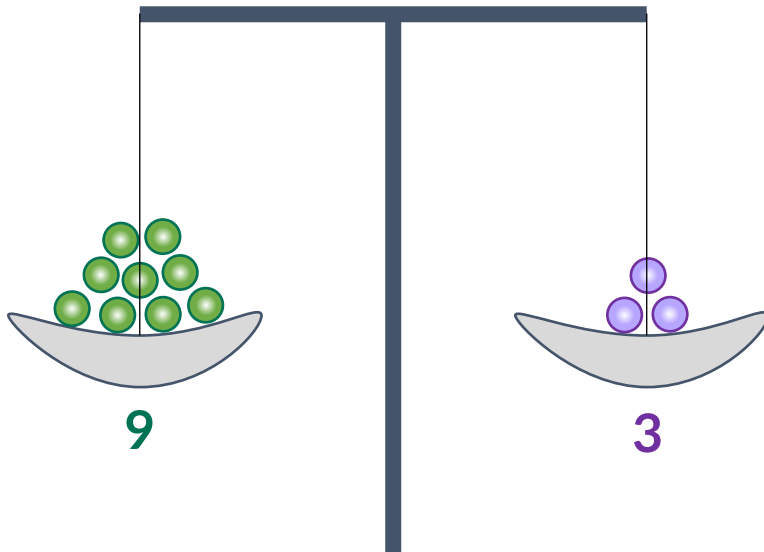
Le Châtelier's Principle

15.6

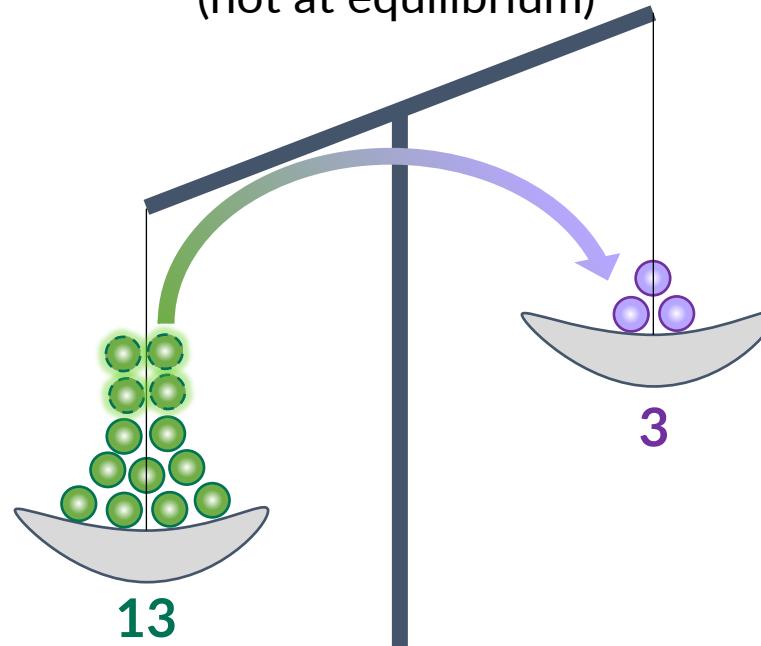
The stress must affect species in expression of K , which defines the equilibrium state, in order for the system to respond. Otherwise, equilibrium unperturbed. For example,



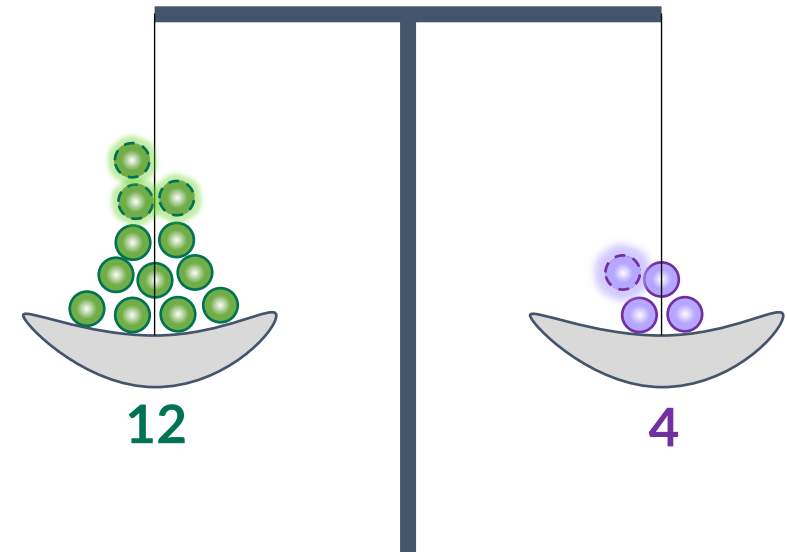
$Q = K$
(at equilibrium)



$Q < K$
(not at equilibrium)



$Q = K$
(at equilibrium)



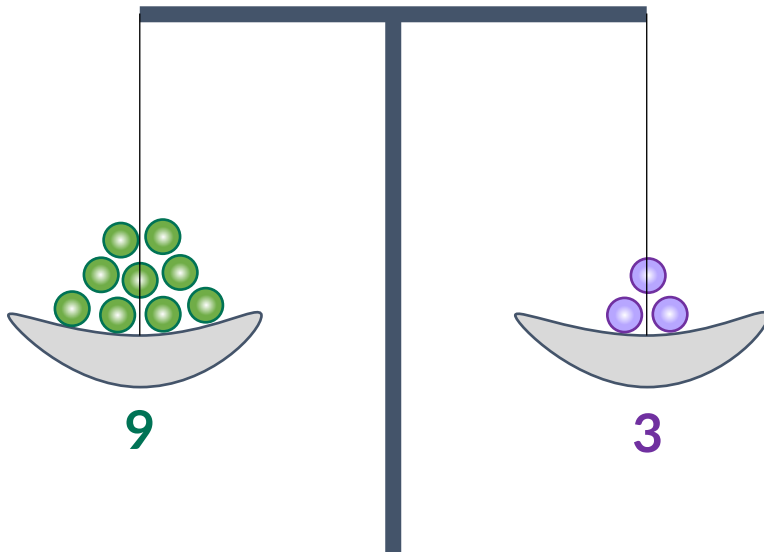
Le Châtelier's Principle

15.6

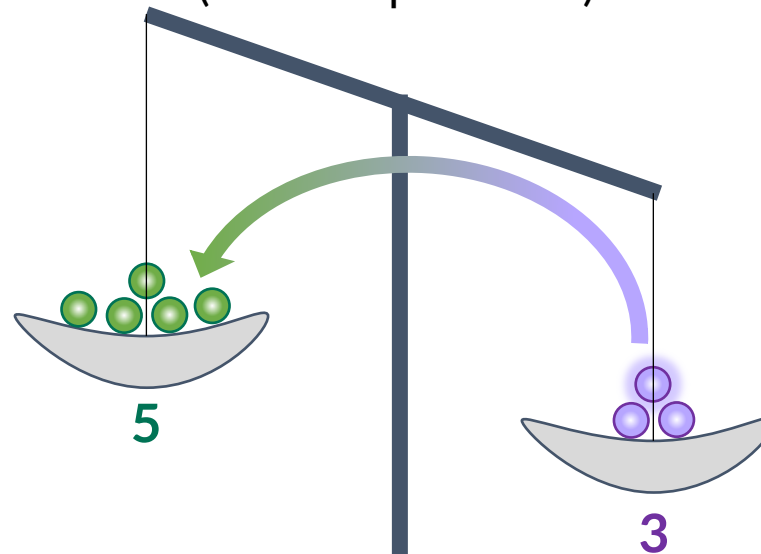
The stress must affect species in expression of K , which defines the equilibrium state, in order for the system to respond. Otherwise, equilibrium unperturbed. For example,



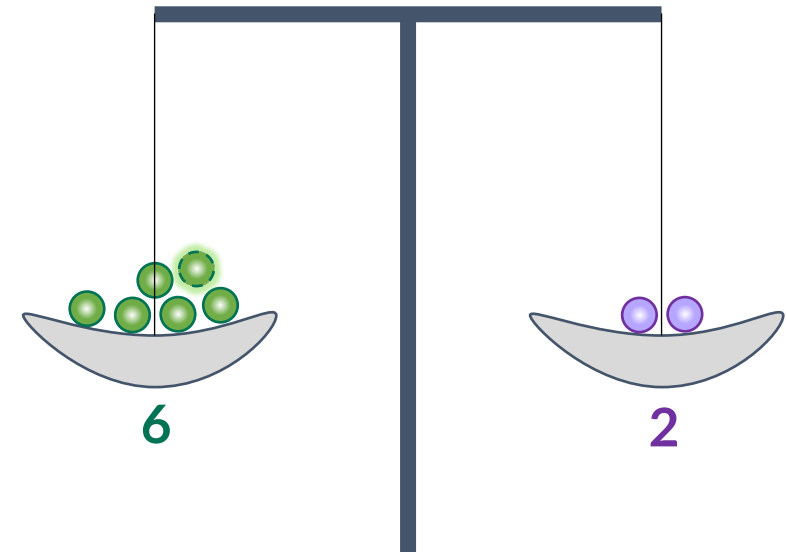
$Q = K$
(at equilibrium)



$Q > K$
(not at equilibrium)



$Q = K$
(at equilibrium)



Le Châtelier's Principle

15.6

Le Châtelier's Principle: Effect of reactant/product quantities

- If a *stress* is applied to a system at equilibrium, the reaction will tend to *shift* in a direction to relieve the stress and re-establish equilibrium.
- Le Châtelier's principle is based on the comparison of Q to K .

	Change in amount	Q vs. K	Reaction shift	Result
Reactant	Increase	$Q < K$		Consumes excess reactant; Forms <i>more products</i>
	Decrease	$Q > K$		Consumes products; Forms <i>more reactants</i>
Product	Increase	$Q > K$		Consumes excess products; Forms <i>more reactants</i>
	Decrease	$Q < K$		Consumes reactant; Forms <i>more products</i>

Le Châtelier's Principle

15.6

Changes in pressure

- If pressure is *increased*, the reaction will shift in a way to *reduce* overall pressure.
- If pressure is *decreased*, the reaction will shift in a way to *increase* overall pressure.

What does this mean in terms of left and right shifts?

- Only consider gaseous species when dealing with pressure changes.
- Look at the balanced chemical equation to determine which side (reactants or products) has more and which has less gases.
- Recall that number of moles, pressures, and volumes of gases are related.
 - a) Shifting to the *side with more gases* increases the pressure.
 - b) Shifting to the *side with less gases* decreases the pressure.

Le Châtelier's Principle

15.6

Changes in pressure

- If pressure is *increased*, the reaction will shift in a way to *reduce* overall pressure.
- If pressure is *decreased*, the reaction will shift in a way to *increase* overall pressure.

Compression → Volume reduced → Pressure increased

- Reaction shifts to side with ***less moles of gas***.

Expansion → Volume increased → Pressure decreased

- Reaction shifts to side with ***more moles of gas***.

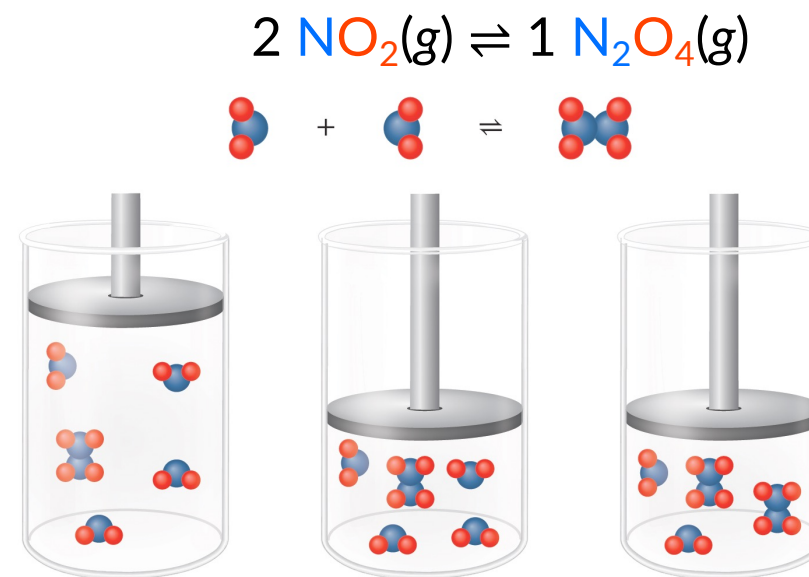
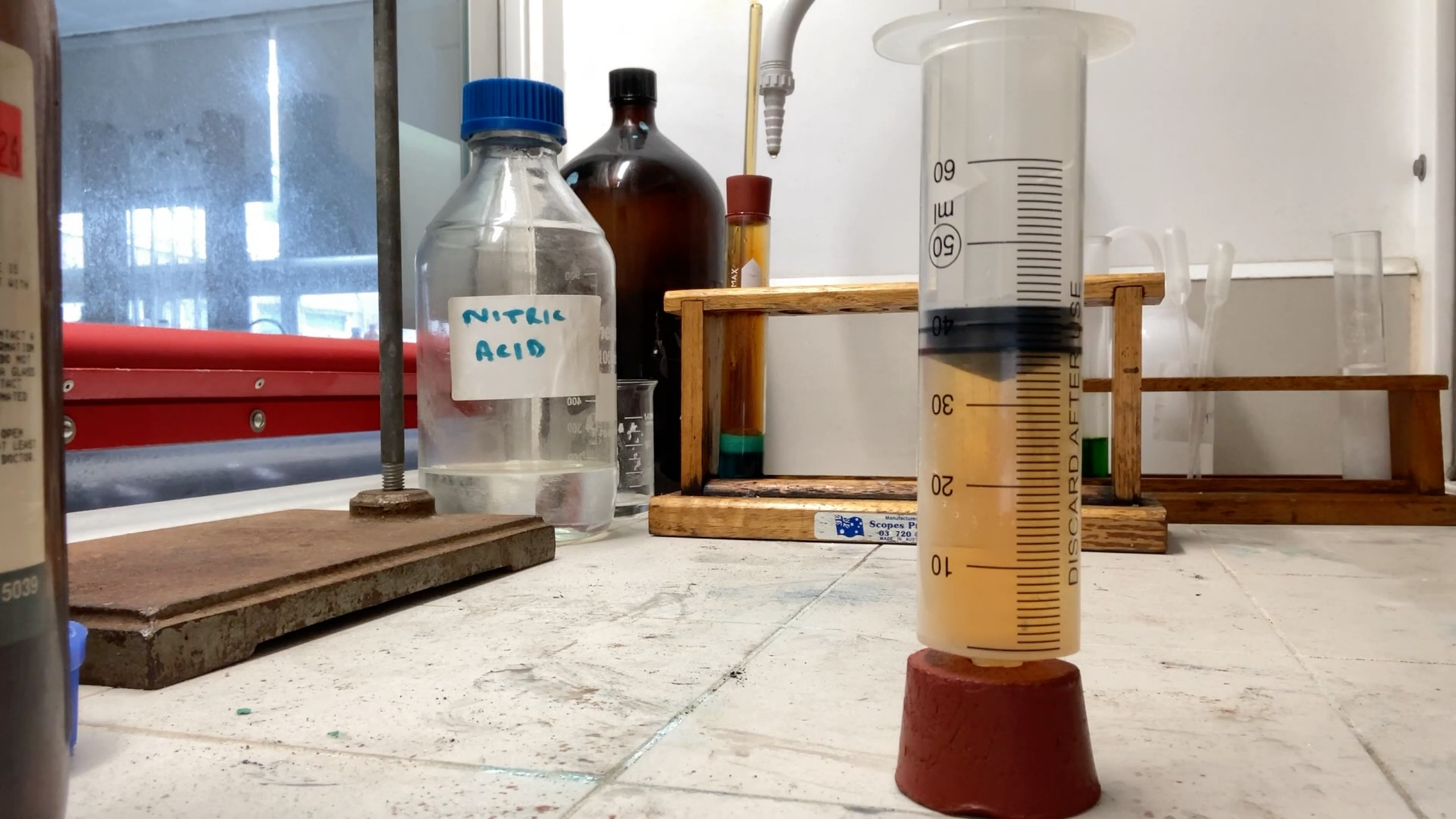


FIGURE 15.7



NITRIC
ACID

60

ml

50

40

30

20

10

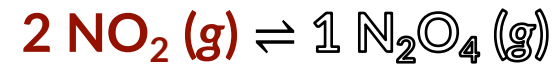
DISCARD AFTER USE

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Le Châtelier's Principle

15.6

What is actually happening?

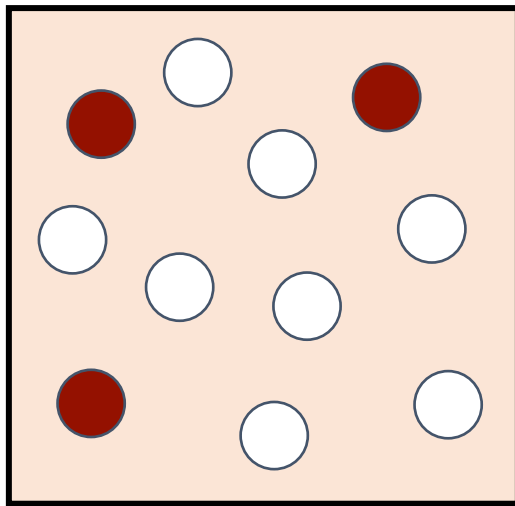


$$K_p = \frac{P_{\text{N}_2\text{O}_4}}{(P_{\text{NO}_2})^2}$$

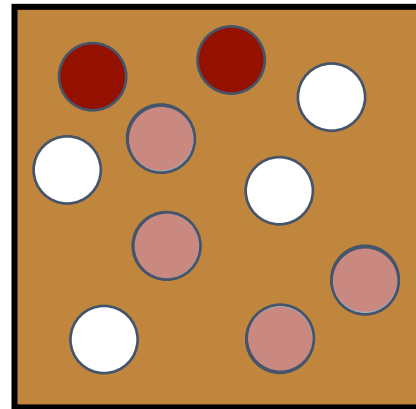
$$P \propto \frac{1}{V}$$

Pressure increased

$$P \propto n$$

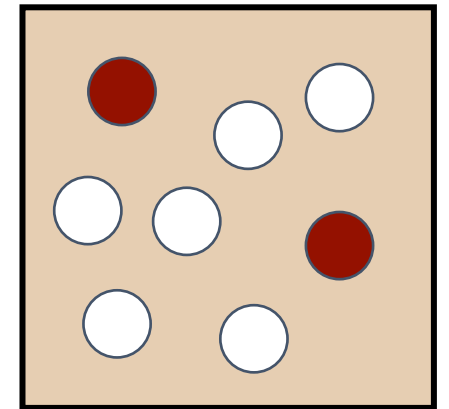


Decrease volume



Shifts right

Lowers pressures by
shifting to side with
less moles of gas



Le Châtelier's Principle

15.6

The effect of pressure and/or volume

- The net change in moles of gas (Δn_{gas}) should be used to determine shifts.

$$\Delta n_{\text{gas}} = n_{\text{gas products}} - n_{\text{gas reactants}}$$

Δn_{gas}	Change in volume	Change in pressure	Reaction shift	Result
$\Delta n_{\text{gas}} < 0$	Expansion			Toward side with <i>more gas</i>
	Compression			Toward side with <i>less gas</i>
$\Delta n_{\text{gas}} = 0$	Either	Either	No shift	No change
$\Delta n_{\text{gas}} > 0$	Expansion			Toward side with <i>more gas</i>
	Compression			Toward side with <i>less gas</i>

IN-CLASS QUESTION 1

Learning Objective(s): 15.8

Select all changes that will cause a shift toward the right for the following reaction?



- ☐ Adding $\text{H}_2\text{O}(g)$
- ☐ Removing some but not all $\text{NaHCO}_3(s)$
- ☐ Adding $\text{Na}_2\text{CO}_3(s)$
- ☐ Removing $\text{CO}_2(g)$
- ☐ Decreasing the overall pressure
- ☐ Decreasing the total volume

What does a catalyst do?

Chemistry in Context

The presence of a catalyst does *not* affect Q or K .

At equilibrium, the following are true:

$$\text{Rate}_{\text{for}} = \text{Rate}_{\text{rev}}$$

$$K = \frac{k_{\text{for}}}{k_{\text{rev}}}$$

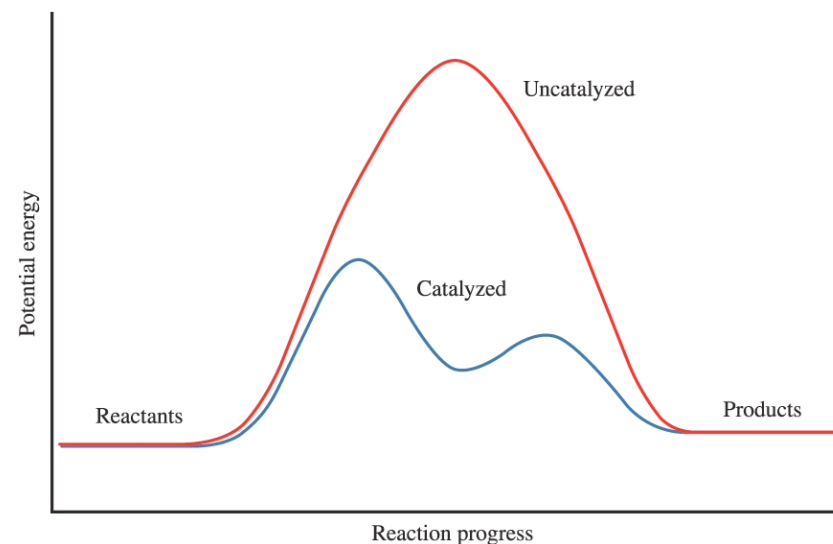


FIGURE 15.11

A catalyst lowers the activation energy for both the forward and reverse reactions *by the same amount*.

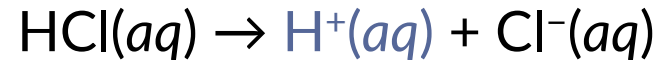
The effect of this is that ***equilibrium is established more quickly*** in the presence of a catalyst, but the equilibrium position itself (defined by K) is *unchanged*.

Ionization Reactions of Acids and Bases

16.1

Arrhenius acid-base theory: The first modern definition of acids and bases reliant on H^+ for acids and OH^- for bases. Only aqueous solutions where water is the solvent.

- Arrhenius acids produce H^+ when dissolved in water.



- Arrhenius bases produce hydroxide ions (OH^-) when dissolved in water.



What chemists call a “proton”?

Chemistry in Context

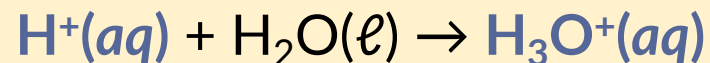
A neutral hydrogen atom (H) has 1 proton, 0 neutrons, and 1 electron.

A cationic hydrogen ion (H^+) has 1 proton, 0 neutrons, and 0 electrons.

- This is why H^+ is referred to, simply, as a “proton.”

Protons (H^+ ions) are *highly reactive* and **do not** exist as H^+ in *aqueous* solutions.

- H^+ will attach to (protonate) a water molecule to form a hydronium ion (H_3O^+).



For our purposes, these are synonymous: proton = H^+ = H_3O^+ = hydronium ion

Brønsted-Lowry Theory

16.2

Brønsted-Lowry acid-base theory: A second definition of acids and bases that is based *only* on proton (hydrogen ions or H^+ ions) behavior.

- Brønsted-Lowry acids (HA) are proton (H^+) *donors*.
- Brønsted-Lowry bases (B) are proton (H^+) *acceptors*.

This is the definition most commonly used today.

Brønsted-Lowry Theory

16.2

A fundamental principle of Brønsted-Lowry theory is that, if a proton donor (an acid, HA) is present then a proton acceptor (a base, B) must also be present.

- Therefore, the reaction between an acid and base is a proton transfer reaction.

Conjugate acid-base pairs differ *only* by the presence or absence of a single proton (H^+).

EXAMPLE PROBLEM

Learning Objective(s): 16.3

For each species, identify the conjugate acid or the conjugate base.

The conjugate acid of HCO_3^-

The conjugate base of HNO_3

The conjugate acid of HPO_4^{2-}

Ionization Reactions of Acids and Bases

16.1

Chemical formulas for acids typically begin with H to indicate that they **ionize** (or dissociate) when dissolved in water.

A **strong acid** is a strong electrolyte and dissociates/ionizes 100 % in water.

- As the ionization is *complete*, this is *not* an equilibrium.
- There are seven strong acids. All other acids are weak acids.

Notes:

1. Some sources consider chloric acid to be a weak acid.
2. Only one of the H atoms completely dissociates. In other words, H_2SO_4 is a strong acid but HSO_4^- is a weak acid.

Strong Acid Name	Formula	Ions in Aqueous Solution	
Hydrochloric acid	HCl	H^+	Cl^-
Hydrobromic acid	HBr	H^+	Br^-
Hydroiodic acid	HI	H^+	I^-
Nitric acid	HNO_3	H^+	NO_3^-
Perchloric acid	HClO_4	H^+	ClO_4^-
Chloric acid ¹	HClO_3	H^+	ClO_3^-
Sulfuric acid ²	H_2SO_4	H^+	HSO_4^-

Ionization Reactions of Acids and Bases

16.1

Chemical formulas for bases typically end with OH (contain hydroxide ions), or end with an H to differentiate them from acids.

A strong base is a strong electrolyte and dissociates/ionizes 100 % in water.

- As the ionization is *complete*, this is *not* an equilibrium.
- There are eight strong bases.
- All other bases are weak bases.

Strong Base Name	Formula	Ions in Aqueous Solution	
Lithium hydroxide	LiOH	Li ⁺	OH ⁻
Sodium hydroxide	NaOH	Na ⁺	OH ⁻
Potassium hydroxide	KOH	K ⁺	OH ⁻
Rubidium hydroxide	RbOH	Rb ⁺	OH ⁻
Cesium hydroxide	CsOH	Cs ⁺	OH ⁻
Calcium hydroxide*	Ca(OH) ₂	Ca ²⁺	OH ⁻ , OH ⁻
Strontium hydroxide*	Sr(OH) ₂	Sa ²⁺	OH ⁻ , OH ⁻
Barium hydroxide*	Ba(OH) ₂	Ba ²⁺	OH ⁻ , OH ⁻

* Note: When Ca(OH)₂, Sr(OH)₂, and Ba(OH)₂ dissociate in water, they yield two OH⁻ anions per unit of strong base.

Ionization Reactions of Acids and Bases

16.1

A weak acid is a weak electrolyte and dissociates/ionizes *incompletely* in water.

- As the ionization is *incomplete* (often <10 %) this is an *equilibrium* process.

- Given the same number of moles of dissolved acid initially, a weak acid will yield significantly less hydronium ions than a strong acid.

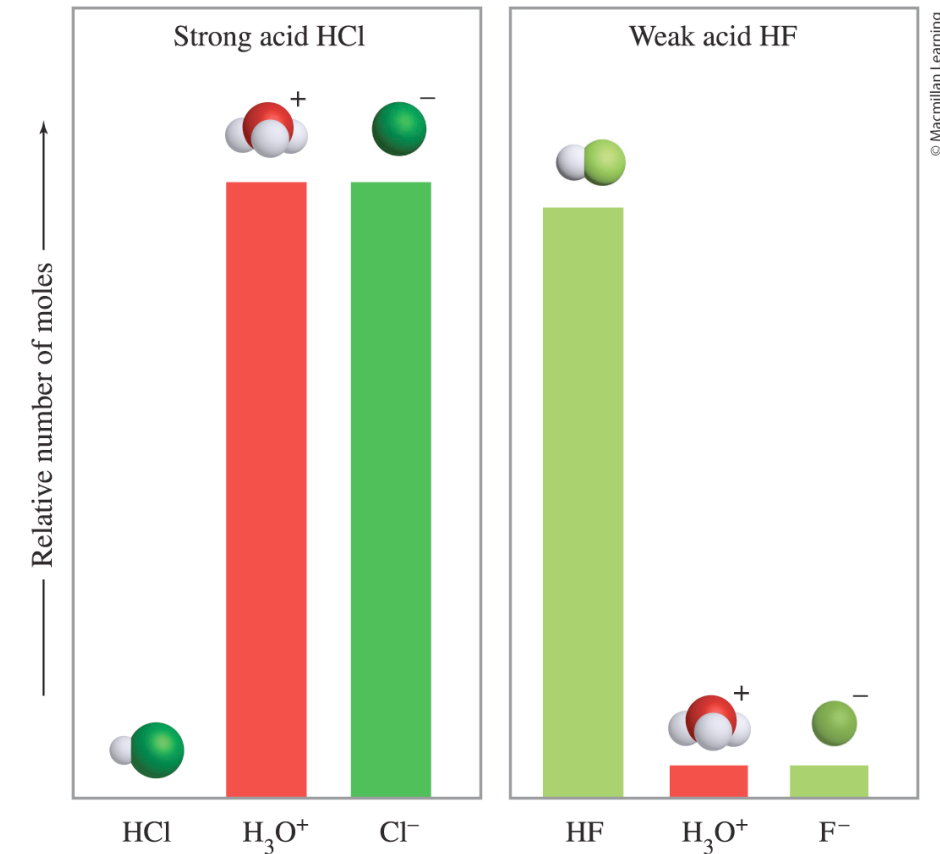


FIGURE 16.1

Ionization Reactions of Acids and Bases

16.1

A weak base is a weak electrolyte and dissociates/ionizes *incompletely* in water.

- As the ionization is *incomplete* (often <10 %) this is an *equilibrium* process.

- Given the same number of moles of dissolved base initially, a weak base will yield significantly less hydroxide ions than a strong base.

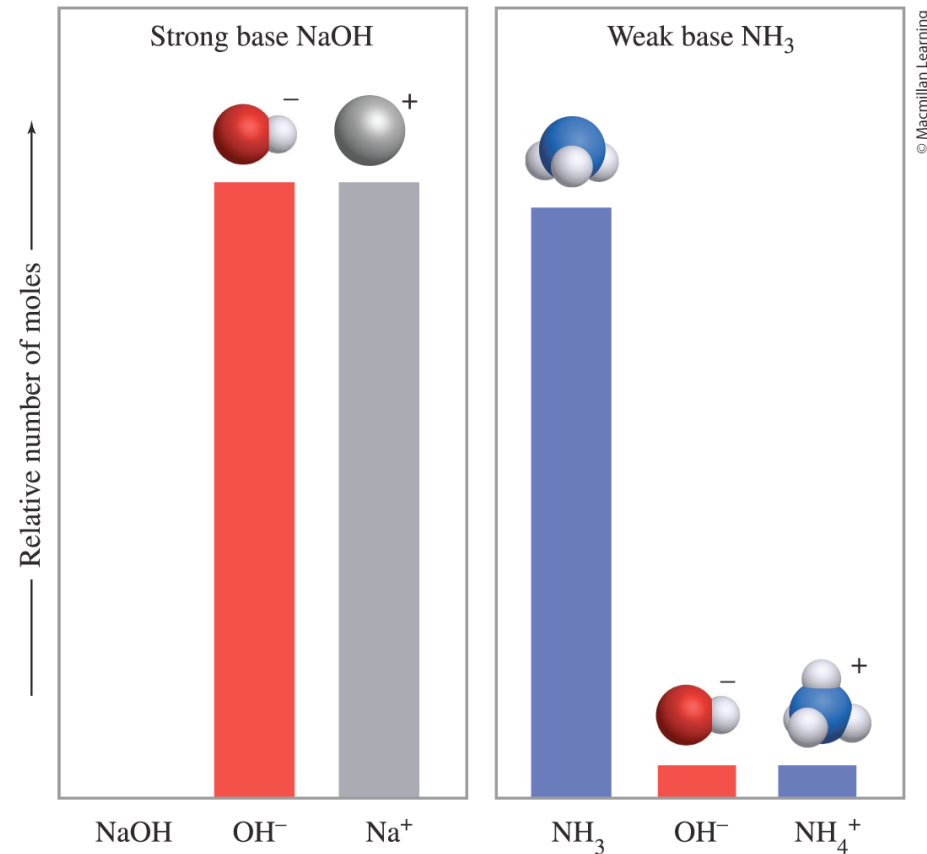


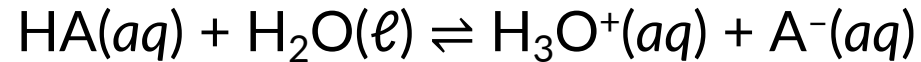
FIGURE 16.2

Weak Acids and Bases

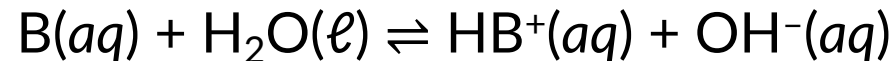
16.5 (abbreviated)

As the ionization of weak acids and weak bases are *equilibria*, the magnitudes of their ionization equilibrium constants (K_a or K_b) is an indication of their relative **strengths**.

Acid ionization constant, K_a



Base ionization constant, K_b



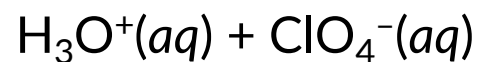
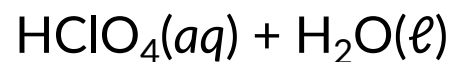
IN-CLASS QUESTION 2

Learning Objective(s): 16.1, 16.4

For the following acids and bases, identify the correct arrow needed to complete each of the ionization reactions in water. Then indicate whether the corresponding values of K_a or K_b are expected to be large or small.

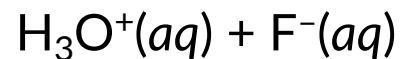
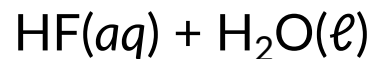
In the box, write $\rightarrow$ or $\rightleftharpoons$ to complete the chemical equation.

Circle one answer per row.



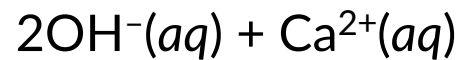
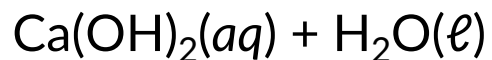
$K_a \gg 1$

$K_a < 1$



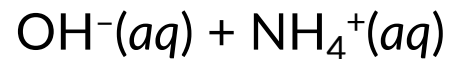
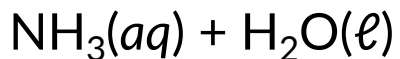
$K_a \gg 1$

$K_a < 1$



$K_b \gg 1$

$K_b < 1$



$K_b \gg 1$

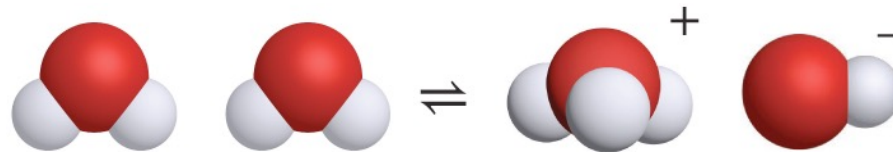
$K_b < 1$

Autoionization of Water

16.3

Water can act as *both* an acid and a base.

- Water molecules can undergo an acid-base reaction with themselves. This is called autoionization.
- The equilibrium constant for the autoionization process is denoted K_w .
- Adding **acids** or **bases** will increase $[H_3O^+]$ or $[OH^-]$, but the system will always re-establish equilibrium, which is defined by the value of K_w .



pH Calculations

16.4

The pH scale is a convenient means of conveying the hydronium ion concentration.

The pOH scale is a convenient means of conveying the hydroxide ion concentration.

- Solutions *often* have pH (or pOH) between 1 and 14, but not always.
- At 25 °C:
 - $\text{pH} < 7$ Acidic solution
 - $\text{pH} = 7$ Neutral solution
 - $\text{pH} > 7$ Basic solution

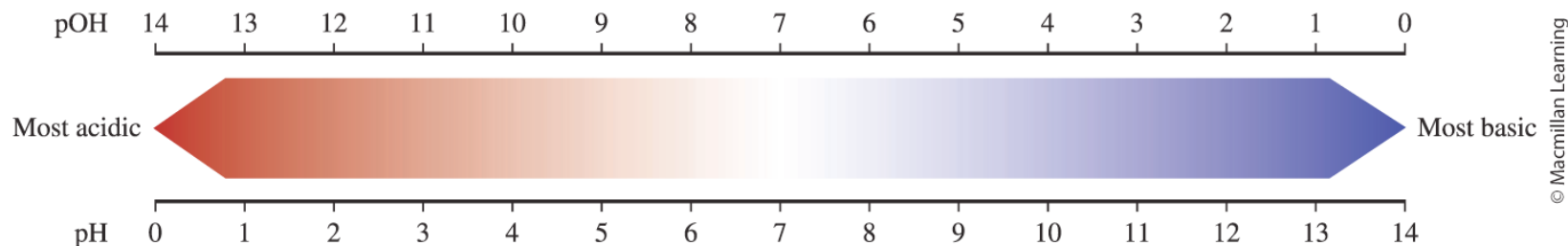


FIGURE 16.4

EXAMPLE PROBLEM

Learning Objective(s): 16.2, 16.5

Consider a 1.57×10^{-3} M $\text{HNO}_3(aq)$ solution at 25 °C.

Calculate the pH, pOH, $[\text{H}_3\text{O}^+]$, and $[\text{OH}^-]$ for this solution.

EXAMPLE PROBLEM

Learning Objective(s): 16.4, 16.5

Order the solutions (of equal concentration) from lowest in pH to highest in pH.

Hint: Answer this question without explicitly calculating the pH values.

A	B	C	D	E
HBr(aq)	KOH(aq)	NH ₃ (aq)	HCN(aq)	Ca(OH) ₂ (aq)

Highest [H ₃ O ⁺]	_____	>	_____	>	_____	>	_____	>	_____	Lowest [H ₃ O ⁺]
Lowest in pH	_____	<	_____	<	_____	<	_____	<	_____	Highest in pH

BONUS QUESTION

Assessed on: Summer 2021, Final Exam

Select all the **true** statements.

- ☐ All initial rates are examples of instantaneous rates.
- ☐ It is possible for molecules to collide with the correct orientation without resulting in a chemical reaction.
- ☐ For a chemical reaction at equilibrium, the amounts of reactants and products are equal.
- ☐ The values of Q and K for a given reaction can never be the same.
- ☐ The majority of strong acid molecules remain intact (i.e., in their protonated forms) in aqueous solution.